5 Years Integrated MSc (Computer Science)
[Course Initiated in 2021-2023]

Semester - VIII

Scheme & Course Structure

Applicable from July 2023

Course Structure with Credits

	Course Type	Subject Names & Credits	Total Credits
Semester-VIII	Core Courses	• MCSCC421-Cloud Computing (4) • MCSCC422-Application Development Project (8)	**4**
	Ability Enhancement Courses		**-**
	Skill Enhancement Courses		**8**
	Discipline Specific Elective Courses	• MCSDE421-Elective-3 (4) • MCSDE422-Elective-4 (4) • MCSDE423-Elective-3 – Practical (2) • MCSDE424-Elective-4 – Practical (2)	**12**
	Generic Elective Courses		**-**
	Semester-VIII (Total Credits)		**24**

TEACHING & EXAMINATION SCHEME FOR 5 YEAR INTEGRATED M. Sc. (COMPUTER SCIENCE) COURSE

Semester-VIII	Course		Hours per week								Credits
	No.	Name	Theory	Practical	Internal	Total	Lectures	Others (Tutorials)	Practical	Toal	
Semester-VIII	MCSCC421	Cloud Computing	70	***	30	100	3	1	***	4	4
	MCSCC422	Application Development Project	***	140	60	200	***	2	12	14	8
	MCSDE421	Elective-3	70	***	30	100	3	1	***	4	4
	MCSDE422	Elective-4	70	***	30	100	3	1	***	4	4
	MCSDE423	Elective-3 - Practical	***	35	15	50	***	***	4	4	2
	MCSDE424	Elective-4 – Practical	***	35	15	50	***	***	4	4	2
		Total	210	210	180	600	9	5	20	34	24

Theory/Tutorial Credits → 1 Hour = 1 Credit, Practical Credits → 2 Hours = 1 Credit

Track wise subjects for Elective-3 and Elective-4			
Track	**Web Technologies**	**Artificial Intelligence & Machine Learning**	**Security**
Subject Code	MCSDE421(1)	MCSDE421(2)	MCSDE421(3)
Subject Name	Web Server Management	Machine Learning	Blockchain Technology
Subject Code	MCSDE423(1)	MCSDE423(2)	MCSDE423(3)
Subject Name	Web Server Management - Practical	Machine Learning - Practical	Blockchain Technology - Practical
Subject Code	MCSDE422(1)	MCSDE422(2)	MCSDE422(3)
Subject Name	Web Data Management	Deep Learning	Internet of Things
Subject Code	MCSDE424(1)	MCSDE424(2)	MCSDE424(3)
Subject Name	Web Data Management - Practical	Deep Learning - Practical	Internet of Things - Practical

Course Name: Cloud Computing

Course Code: MCSCC421

Objectives:

At the end of the course, student will be able to

- Articulate the main concepts, key technologies, strengths, and limitations of cloud computing and the possible applications for state-of-the-art cloud computing
- Implement business solutions over cloud computing platform
- Harness the cloud infrastructure to provide efficient software-based solutions
- To understand the service model with reference to cloud computing

Pre-requisites:

Basic Knowledge of Operating Systems, Knowledge of Object-Oriented Programming Language, Knowledge of Database Systems

Contents:

1. **Cloud Computing Fundamentals**
   Evolution and enabling technologies for cloud computing, Motivation for cloud computing, Benefits & challenges, Defining cloud computing, Principles and challenges, Cloud computing model, Cloud computing services

2. **Virtualization, Scaling, Capacity Planning & Cloud Technologies**
   Resource Virtualization, Approaches to virtualization, Hypervisors, Resource pooling, Resource Sharing and Provisioning, Scaling in the cloud, Capacity Planning and Load Balancing
   The File system and Storage services, Database technologies, Networking technologies over cloud & CDN, Cloud Management and Programming Model, Security reference model, Security aspects, Platform related security, Audit and Compliance, Portability and Interoperability issues

3. **Open Source support for cloud**
   Open source tools for IaaS, Open source tools for PaaS, Open source tools for SaaS, Open source tools for research, Distributed Computing Tools for Management of Distributed Systems

4. **Software development in the cloud**
   Introduction, Different perspectives on SaaS development, Challenges in cloud environment for software development, Launching a virtual machine, Creation of an instance, Connecting to an instance, Security services & roles, Storage Services, Database services and launching the database instance, Creating a static website, Deployment of code, Harnessing Serverless Architecture, Container Services, Notification Services

References:

1. Bahga A., Madisetti V., “Cloud Computing – A Handson Approach”, University Press
2. Bhowmik S., "Cloud Computing", Cambridge University Press
3. Baron J., et al., "AWS Certified Solutions Architect Official Study Guide: Associate Exam", Sybex
4. Hunter T., Porter S., "Google Cloud Platform for Developers: Build highly scalable cloud solutions with the power of Google Cloud Platform", Packt Publishing Limited
5. Chandrasekaran K., "Essentials of Cloud Computing", CRC Press
6. Marinescu D., "Cloud Computing - Theory and Practice", Morgan Kauffman
7. Sosinsky B., “Cloud Computing Bible”, Wiley India
8. Buyya R., Broberg J., Goscinski A., “Cloud Computing : Principles and Paradigm”, John Wiley & Sons

Accomplishments of the student after completing the Course:
At the end of the course, students will be able to

- Decide and implement appropriate cloud computing infrastructure for any organization
- Design customized programming solutions using cloud computing model

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Course Name: Application Development Project

Course Code: MCSSE421

Objectives:

The aim of this course is

- To introduce structured development of software systems
- To acquaint students to various techniques of requirements determination
- To introduce the concepts of analysis and design
- To model the system with various software diagrams
- To develop a complete system/module of a large system using software engineering concepts
- To prepare document/report of the system

Prerequisites:

Knowledge of Programming

Contents:

1. **Introduction to System Analysis & Design**
   Concepts of System Analysis & Design, Components of a System, Users of a System, System Development Life Cycle, Role of System Analyst, Various approaches to develop a system, Using a "systems" approach to build whole system

2. **Structured Systems Development Approach**
   Requirements Gathering, surveying existing systems, Performing Feasibility Study before building a system, Formulating SRS, Designing Data Flow Diagrams, Entity-Relationship Diagrams, Designing Data Dictionary, Designing effective Input and Reports, Implementing EOD, EOM and EOY Procedures, Documenting the system

Guidelines for Project:

1. Projects can be created by a single student or in a group of 2-3 students.
2. The definition of project is to be submitted within 15 days of the starting of the semester.
3. The project should be free from plagiarism.
4. A student/group is required to report to their project guides on regular basis.
5. A student/group is required to present the project during internal exams.
6. It is advisable that the project must include different forms and reports.
7. It is necessary to prepare system diagrams, data dictionary as per the convention.
8. Project documentation should be prepared for the semester end examination.
9. In the semester end examination, the project should be presented to the examiners along with the documentation (report).

References:

1. Pressman R., Maxim B., "Software Engineering – A practioner's approach", McGraw Hill
2. Kendall & Kendall. "System Analysis and Design ", Prentice Hall India (PHI)

3. James Sen, “ Analysis and Design of Information System”, PHI
4. Arthur Langer, “ Analysis and Design of Information Systems “, Springer
5. Jeff Johnson, “ Designing with the Mind in Mind “, Morgan Kaufmann
6. Dennis, Wixom, Roth “System Analysis and Design”, Wiley India
7. Joseph Valacich, Joey F. George, “Modern Systems Analysis and Design”, Pearson
8. Priti Srinivas Sajja., “Essence of System Analysis and Design”, Springer

Accomplishments of the student after completing the Course:

After completion of this course, students will be able to

- Analyse and Design a system based on user requirements
- Develop a software system using software engineering concepts
- Prepare a document for the system

Subject Name: Web Server Management

Subject Code: MCSDE421 (1)

Objectives:
The aim of this course is

- To enable students to know the concept of web server management and apply the concepts in practical

Prerequisite:
Basics of Linux

Contents:

1. **Installation, Adding Common Modules**
   Install Apache on Linux and Windows, Debian Packages, Subversion Sources, Apache Toolbox, Starting and stopping Apche, Restarting Apache, Apache uninstallation, Which Apache versions Version of Apache to Use, upgrades, config.nice, boot, Useful configure script, options configure Options, Finding Apache's files location Files, Installing a Generic installation modules, third-party modules, Installing Unix mod_dav, Installing Windows mod_dav, Installing installation mod_perl, Unix Unix mod_perl, Installing Unix mod_php, Installing Windows mod_php, Installing installation mod_ssl, finding Modules Using modules locating, modules.apache.org, Installing mod_security,

2. **Logging, Virtual Hosts, Aliases, Redirecting, and Rewriting**
   Log entries, logs error, POST Contents, IP addresses and clients, MAC addresses, Cookies, Local Pages, logs rotating, Hostnames, Maintaining logs virtual hosts, Proxy Requests, IP addresses servers, Referring Page, browser, software name, logs arbitrary request, logs arbitrary response, MySQL database logs, Syslog, User Directories, Setting Up Name-Based Virtual Hosts, default virtual hosts, address-based virtual hosts, virtual hosts address-based mixing with name-based, Mass Virtual Hosting with mod_vhost_alias, Mass Virtual Hosting Using Rewrite Rules, Logging for virtual hosts, Logfile, port-based virtual hosts, Displaying the Same Content on Several Addresses, Virtual Hosts in a Database, Directories URLs mapping, Creating a URLS, URLs aliasing, directories URLs mapping to same CGI directory, CGI directories creating, Redirecting URLs to another location, Permitting case-insensitive URLs, Showing PHP source, URLs (Uniform Resource Locators) replacing text, CGI arguments, requests unreferred denying access, query strings, SSL (Secure Socket Layers) redirecting, directories turning into hostnames, redirecting to single host, arguments, URLs (Uniform Resource Locators) elements, directories rewriting, query arguments, Using AliasMatch

3. **Security, SSL & Dynamic Content**
   Authentication system, Single-Use Passwords, Expiring Passwords, Limiting Upload Size, Images from Being Used Off-Site, requiring Strong Authentication, .htpasswd Files, Digest Authentication, Security in a Subdirectory, Lifting Restrictions, Selectively, files ownership, MySQL databases User Credentials, authentication usernames, authentication passwords, brute-force password attacks, Digest authentication, Credentials Embedded in URLs, WebDAV security, WebDAV enabling, proxies URL

access, files wrappers, module sets, Web Root, Restricting Range Requests, DoS attacks, mod_security, Chrooting Apache with mod_security, authentication migrating, Mixing read-only access, Using redirecting permanent, Installing SSL, Windows SSL installation, SSL Certificates , SSL (Secure Socket Layers) CAs, Serving a SSL, Authenticating with Client authentication SSL, virtual hosts SSL, SSL certificates wild card, Enabling a CGI Directory, CGI scripts enabling, CGI directories default documents, Using CGI programs launching, Using CGI scripts extensions, CGI testing setup, CGI form parameters, SSIs (Server-Side Includes) , Displaying date, last modified date, Including a Standard Header, Including the CGI program output, CGI scripts, CPAN, mod_perl Handler, PHP script handling, PHP installation, CGI parsing output, Parsing SSIs, Getting Perl scripts, Enabling Python Script Handling

4. Error Handling, Proxies, Performance & Directory Listings

A error handling host fields, CGI Scripts , Customized Error Messages, Error Documents in Multiple Languages , error handling redirecting, Internet Explorer Display Your Error Page , Notification on Error Conditions, Securing Your Proxy Server, Open Mail Relay, Forwarding Requests to Another Server, Blocking Proxied Requests, mod_perl proxying content, caching proxy servers, Filtering Proxied Content, authentication Proxied Server, proxies, mod_proxy_balancer, proxies virtual hosts, FTP, How Much Memory You Need, Benchmarking Apache with ab, KeepAlive Settings, Getting a Snapshot of Your Site's Activity, DNS lookups, Symbolic Links, .htaccess Files, Content Negotiation, Optimizing Process Creation, Thread Creation, Caching, load balancing, Caching Directory Listings, Speeding Up mod_perl, Dynamic Content, Directory/Folder Listings, Header and Footer on Directory Listings, Applying a Style sheet, Hiding Things from the Listing, Searching for Certain Files in a Directory Listing, directories sorting, Client-Specified Sort Order, List directories formatting, client-specified formatting, Adding Descriptions to Files, Auto generated Document Titles, Changing the icons, Listing the Directories First, files version number, Allowing the End User to Specify Version Sorting, User Control of Output, End User to Modify the Listing, Suppressing Certain Columns, Showing Forbidden Files, directories aliases

References:

1. Rich Bowen, Ken Coar, “Apache Cookbook”, O’Reilly
2. Ed Sawicki , “Guide to Apache” , Cengage
3. Ivan Ristic, “Apache Security”, O’Reilly
4. Steve Silva, “Web Server Administration”, Cengage
5. Rosemary Scoular, “Apache: The Definitive Guide”, O’reilly

Accomplishments of the student after completing the course:

After completion of this course student will be able to

- Manage web server and use it effectively
- Apply security settings to web server

Course Name: Web Data Management

Course Code: MCSDE422 (1)

Objectives:

The aim of this course is to

- Articulate the main concepts, key technologies and the possible applications for web data management
- Implement business solutions over cross platform computing
- Harness the searching & indexing technologies for data management
- Understand the role of web analytics for Internet traffic

Pre-requisites:

Basic Knowledge of Operating Systems, Knowledge of Object-Oriented Programming Language, Knowledge of Database Systems

Contents:

1. **Modeling Web Data**
   XML, Web Data Management with XML, XML dialects and standards, XPath and XQuery Basics, Working with XPath, FLWOR expressions in XQuery, XPath Foundations, Typing, Typing, Automata on Words & Trees, Schema Languages for XML, Typing Graph Data, XML Query Evaluation, Managing an XML Database, Tree Pattern Evaluation using SAX

2. **Web Data Semantics and Integration**
   Ontologies, RDF and OWL, Querying Data through Ontologies, Data Integration, Wrappers and Data Extraction, Ontologies in Practice, Mashups with Yahoo! Pipes and XProc

3. **Building Web Scale Applications**
   Web Search, Parsing the Web, Web Information Retrieval, Overview of Distributed Systems, Distributed Access Structures, Distributed Computing, Full-Text Indexing, Using Recommendations, Large-Scale Data Management, Using Semi-Structured Database

4. **Web Analytics**
   Why Understanding Your Web Traffic is Important to Your Business - Available Methodologies and Their Accuracy , Page Tags and Log Files, Apply key concepts of Web Analytics, Introduction to Web Analytics Apply key tools and diagnostics associated with Web analytics, Google Analytics Features, Benefits and Limitations , Open Source Tool , Tracking the Mobile Visitor, Use free open source Web analytics tools to collect, identify information and data Prepare Embedded JavaScript Page Tracking Code, Reports Explained , Web Analytics, Up and Running , Traffic Sources: AdWords, Implement website traffic reports

References:

1. Abiteboul S., et. Al., “Web Data Management”, Cambridge University Press
   (Online: http://webdam.inria.fr/Jorge/files/wdm.pdf)
2. Brian Clifton, Advance Web Metrics with Google Analytics, Sybex
3. Avinash Kaushik., “Web Analytics 2.0, The Art of Online Accountability and Science of Customer” , Wiley
4. Michael Beasley, “Practical Web Analytics for User Experience”, Morgan Kaufmann
5. Stephan A. Miller, Piwik “Web Analytics Essentials”, Packt
6. Jim Sterne , “Web Metrics: Proven Methods for Measuring Web Site Success”, Wiley

Accomplishments of the student after completing the Course:
At the end of the course, students will be able to

- Decide and implement appropriate data model for WWW
- Design customized search engines
- Perform Web Analytics

Course Name: Machine Learning

Course Code: MCSDE421 (2)

Objectives:

The main objective of this course is to enable students to

- Develop a strong theoretical foundation for understanding of state of art Machine Learning Algorithms
- Identify, formulate and solve machine learning problems that arise in practical applications
- Explore various tools and technologies that enable Machine Learning

Prerequisites:

Mathematical Foundations, Programming in C++/Python

Contents:

1. **Overview of Machine Learning**
   Introduction to Machine learning from data, Types of Machine Learning: Supervised, Unsupervised, Reinforcement, concepts of regression, classification, clustering

2. **Linear Regression & Logistic Regression**
   Scatter diagram, Model representation for single variable, Single variable Cost Function, Least Square line fit, Normal Equations, Gradient Descent method for Linear Regression, Assumptions in linear regression, properties of regression line, Model Performance through R^2, Multivariable model representation, Multivariable cost function, multiple linear regression, Normal Equations and non-invertibility , Gradient Descent method for multiple linear regression , Overfitting, Underfitting, Bias and variance , Regularization, Issues of using Linear Regression in Classification, Sigmoid function, odds of an event, Logit function, Decision Boundary, Maximum Likelihood function, Linear regression verses Logistic Regression, Cost function, Multi-classification , confusion matrix, statistical measures to measure binary classification : Recall, sensitivity, specificity, precision, accuracy, pros and cons of logistic regression

3. **Supervised Learning**
   Classification problems; decision boundaries; K nearest neighbour methods, Linear classifiers, Bayes' Rule and Naive Bayes Model, SVM - Introduction, Support Vectors & Margin, Optimization Objective, Linear & Non-Linear SVM, Hard Margin & Soft Margin in, Large Margin Classifiers, Kernels, SVM practical considerations, Ensemble methods for classification and regression: Bagging, Random Forests, Boosting, Decision Tree

4. **Unsupervised learning**
   Cluster Analysis, Classification and Clustering , Definition of Clusters ,Clustering Applications , Distance measures, Proximity Measures for Discrete Variables, Proximity Measures for Mixed Variables, Partitional Clustering, Clustering Criteria, K-Means

Algorithm, Fuzzy Clustering , Hierarchical Clustering, Agglomerative Hierarchical Clustering, Divisive Hierarchical Clustering, Cluster Validity, External Criteria, Internal Criteria

References:

1. Mitchell T., “Machine Learning”, McGraw Hill Publications
2. Grus J,. “Data Science from Scratch”, O’Reilly Publications
3. Alpaydin E., “Introduction to Machine Learning”, MIT Press
4. Coelho R., “Building Machine Learning Systems with Python”, Packt Publishing Ltd.
5. Marsland S., “MACHINE LEARNING: An Algorithmic Perspective, CRC Press
6. Wunsch D,. Xu R,. “Clustering”, IEEE Press
7. “Introduction to Machine Learning”; Ethem Alpaydın; The MIT Press

Accomplishments of the student after completing the course:

After completion of the course, students should be able to

- Develop an appreciation for what is involved in learning models from data
- Understand a wide variety of learning algorithms
- Understand how to evaluate models generated from data
- Understand and develop application involving computer vision and Natural Language Processing

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Course Name: Deep Learning

Course Code: MCSDE422 (2)

Objectives:
The aim of this course is to enable students to

- Introduce with the latest algorithms and architectures of deep learning with practical viewpoint
- The necessary background to understand the ongoing research and gain required implementation knowledge

Prerequisites:
Basics of Linear Algebra, Calculus, Probability and Machine Learning Fundamentals

Contents:

1. **Feed-forward Deep Networks**
   Overview of Deep Networks, Bias and Variance, the curse of dimensionality, Vanilla MLP, Flow Graphs and Back propagation, Universal Approximation Theorem, Feature representation

2. **Convolutional Networks, Recurrent and Recursive Nets**
   Concept of Convolution, Convolution Operation, Pooling, Stride, Convolution Modules, Efficient Convolution Algorithms, Unfolding Flow graphs and Parameter Sharing, Recurrent Neural Networks, Bidirectional RNNs, Deep Recurrent Architecture, Recursive Neural Networks, Auto-Regressive Networks, Recurrent Vs. Recursive Neural Nets
3. **Regularization of Deep Models**
   Importance of Regularization, L1 and L2 Regularization, Parameter Norm Penalty, Regularization as Constrained Optimization, Weight Regularization, Classical Regularization as Noise Robustness, Dropout, Drop Connect, Multi-Task Learning, Adversarial Training

4. **Optimization for Training Deep Models & Implementations**
   Need of Optimization for Model Training, Challenges in Optimization, Basic Algorithms, Adaptive Learning, Second order methods, Natural gradient methods, Global Optimization, Image Classification, Types of Image Classifiers, Building a deep neural architecture, Building feature set, Preparing Data for training, Adding Dropout, Understanding Alex Net and Google LeNet, Using existing architectures in applications, Using inbuilt Tensor Flow functionality to build a Convolutional Neural Network

References:

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, “Deep Learning”, MIT Press

2. Jeff Heaton, “Artificial Intelligence for Humans: Deep Learning and Neural Networks Book – 3”, Heaton Research Inc.
3. Giancarlo Zaccone, “Deep Learning with Tensor Flow”, Packt Publishing Ltd.

Accomplishments of the student after completing the Course:

At the end of the work students will be able to

- Understand various algorithms for deep learning
- Understand the concepts and architecture related to deep learning
- Design models to solve real life challenges using deep learning

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Course Name: Blockchain Technology

Course Code: MCSDE421 (3)

Objectives:

The aim of this course is

- To introduce block chain technology and Cryptocurrency to the students
- To make students to integrate ideas from blockchain technology into their own projects

Prerequisites:

Distributed Databases

Contents:

1. **Blockchain Fundamentals & Cryptocurrency**
   Overview of Cryptography, Introduction to Blockchain, Components, Concept of Block, Block Chain Types and Consensus Mechanism, Cryptocurrency Basics, Types of Cryptocurrency, Cryptocurrency Usage : Ecosystem Players, Cryptomining, Airdrop, Token and Coin Burning, Investing and Trading, Cryptocurrency Safety

2. **Public and Private Blockchain System**
   Introduction, Popular Public Blockchain, Bitcoin, Ethereum, Smart Contract, Characteristics of Private Blockchain, Private Blockchain and Open source

3. **Consortium Blockchain and ICO**
   Introduction, Characteristics, Hyperledger Platform, Fundraising Methods, Lunching ICO, Investing in ICO, ICO Pros and Cons, Evolution of ICO, ICO Platforms : Launching and Listing

4. **Security & Applications of Blockchain**
   Security Aspect in Bitcoin, Challenges, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract, Uses of Blockchain in Computing, Limitations and Challenges in Blockchain, Blockchain Platform using Python

References:

1. Chandramouli Subramanian, Asha A George, Abhilash K A and Meena Karthikeyan, “Blockchain Technology”, Universities Press
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, “Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction”, Princeton University Press
3. Melanie Swan, “Blockchain Blue print for Economy”
4. Daniel Drescher, “Blockchain Basics: A Non-Technical Introduction in 25 Steps”
5. Imran Bashir, “Mastering Blockchain”, 2/E, Packt publishing, Mumbai

Accomplishments of the student after completing the Course:

At the end of the course, students will be able to

- Understand Blockchain Mechanism
- Understand Various Applications of Blockchain

Course Name: Internet of Things

Course Code: MCSDE422(3)

Objectives:

The aim of this course is to enable students to

- Understand general concepts of Internet of Things (IoT)
- Recognize various devices, sensors and applications
- Apply design concept to IoT solutions
- Analyze various M2M and IoT architectures
- Evaluate design issues in IoT applications
- Create IoT solutions using sensors, actuators and Devices

Prerequisites:

Basics of Wireless networking, Network Security

Contents:

1. **Introduction to IoT**
   Sensing, Actuation, Networking basics, Communication Protocols, Sensor Networks, Machine-to-Machine Communications, IoT Definition, Characteristics. IoT Functional Blocks, Physical design of IoT, Logical design of IoT, Communication Protocols, IoT enabling technologies, Sensor Networks, Machine-to-Machine Communications

2. **IoT & M2M**
   Difference between IoT and M2M, IoT architecture, Software define Network, SDN for IoT, Data Handling and Analytics, Cloud Computing, Sensor-Cloud, Fog Computing

3. **IoT Architecture**
   Introduction, State of the art, Architecture Reference Model- Introduction, Reference Model and architecture, IoT reference Model, IoT Reference Architecture- Introduction, Functional View, Information View, Deployment and Operational View, Other Relevant architectural views

4. **IOT Applications and Challenges**
   Case studies: Lighting as a service, Intelligent Traffic systems, Smart Parking, Smart water management, IoT for smart cities, *IoT in Indian Scenario*: IoT and Aadhaar IoT for health services. IoT for financial inclusion IoT for rural empowerment

   Challenges in IoT implementation: Big Data Management, Connectivity challenges, Mission critical applications, security and privacy issues

References:

1. Vijay Madisetti and Arshdeep Bahga, "*Internet of Things (A Hands-on-Approach)*" ,1st edition, Orient Blackswan, Universities Press
2. Francis daCosta, "*Rethinking the Internet of Things: A Scalable Approach to Connecting everything*", 1st edition, Apress Publications

3. Cuno Pfister, "*Getting Started with the Internet of Things*", O'Reilly Media, 2011, ISBN: 978-1-4493-357-1

Accomplishments of the student after completing the Course:

At the end of the work students will be able to

- Apply IoT concepts and IoT Standards
- Understand Components and relevance of IoT System for the future
- Build IoT Applications
- Apply IoT in smart city environment in Indian scenario
- Analyze challenges in IoT implementation

Subject Name: Web Server Management - Practical

Subject Code: MCSDE423(1)

Objectives:

To enable students to know

- Install & Configure Web Server
- Setup Web Server for Security
- Optimize Web Server for better Performance

Contents:

1. **Setting up Apache Web Server**

   Install Apache on Linux and Windows, Debian Packages, Subversion Sources, Apache Toolbox, Starting and stopping Apache, Restarting Apache, Apache uninstallation, Finding Apache's files location Files

   Installing a Generic installation modules, third-party modules, Installing Unix mod_dav, Installing Windows mod_dav, Installing installation mod_perl, Unix Unix mod_perl, Installing Unix mod_php, Installing Windows mod_php ,Installing installation mod_ssl, finding Modules Using modules locating, modules.apache.org, Installing mod_security,

   Setting Up Name-Based Virtual Hosts, default virtual hosts, address-based virtual hosts, virtual hosts address-based mixing with name-based, Mass Virtual Hosting with mod_vhost_alias, Mass Virtual Hosting Using Rewrite Rules, Logging for virtual hosts, Logfile, port-based virtual hosts, Displaying the Same Content on Several Addresses, Virtual Hosts in a Database

   Dealing with Logs, Syslog and User Directories, Using CGI, Dealing with URLs

2. **Applying Security, Dealing with Proxies & Dynamic Content:**

   Using .htpasswd Files, Digest Authentication, Security in a Subdirectory, Lifting Restrictions, Selectively, files ownership, MySQL databases User Credentials, authentication usernames, authentication passwords, brute-force password attacks, Digest authentication, Credentials Embedded in URLs, WebDAV security, WebDAV enabling, proxies URL access, files wrappers, module sets, Web Root, Restricting Range Requests, DoS attacks, mod_security, Chrooting Apache with mod_security, authentication migrating, Mixing read-only access, Using redirecting permanent

   Installing SSL, Windows SSL installation, SSL Certificates , SSL (Secure Socket Layers) CAs, Serving a SSL, Authenticating with Client authentication SSL, virtual hosts SSL, SSL certificates wild card

   Securing Your Proxy Server, Open Mail Relay, Forwarding Requests to Another Server, Blocking Proxied Requests, mod_perl proxying content, caching proxy servers, Filtering Proxied Content, authentication Proxied Server, proxies, mod_proxy_balancer, proxies virtual hosts, FTP

Enabling a CGI Directory, CGI scripts enabling, CGI directories default documents, Using CGI programs launching, Using CGI scripts extensions, CGI testing setup, CGI form parameters, SSIs (Server-Side Includes) , Displaying date, last modified date, Including a Standard Header, Including the CGI program output, CGI scripts, CPAN, mod_perl Handler, PHP script handling, PHP installation, CGI parsing output, Parsing SSIs, Getting Perl scripts, Enabling Python Script Handling

References:

1) Rich Bowen, Ken Coar, “Apache Cookbook”, O’Reilly
2) Ed Sawicki , “Guide to Apache” , Cengage
3) Ivan Ristic, “Apache Security”, O’Reilly
4) Steve Silva, “Web Server Administration”, Cengage
5) Rosemary Scoular, “Apache: The Definitive Guide”, Oreilly

Accomplishments of the student after completing the course:

After completion of this course student will be able to know

- To manage web server and use it effectively.

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Course Name: Web Data Management - Practical

Course Code: MCSDE424(1)

Objectives:

At the end of the course, student will be able to

- Implement Web Data Modeling & Integration of Web Data
- Implement business solutions over cross platform computing
- Implement the searching & indexing technologies for data management
- Using web analytics for Internet traffic

Pre-requisites:

Basic Knowledge of Operating Systems, Knowledge of Object-Oriented Programming Language, Knowledge of Database Systems

Contents:

1. **Modeling Web Data & Web Data Semantics Integration**

   EXIST: Installing EXIST, Getting started with EXIST, Running XPath and XQuery queries with the sandbox, Programming with EXIST, Projects.

   Tree Pattern Evaluation using SAX

   Wrappers & Data Extraction with XSLT, Working with YAGO, Mashups with Yahoo! Pipes and XProc

2. **Web Scale Applications**

   Overview of LUCENE, Installing and configuring LUCENE, Indexing plain-text, Tuning the scoring

   Installing and running HADOOP, Running MAPREDUCE jobs, Running PIGLATIN scripts, Running in cluster mode

   Using COUCHDB, Dealing with data in COUCHDB, Using JSON Semi-structured data

   Using Google Analytics

References:

1. Abiteboul S., et. Al., “Web Data Management”, Cambridge University Press (Online: http://webdam.inria.fr/Jorge/files/wdm.pdf)
2. Brian Clifton, Advance Web Metrics with Google Analytics, Sybex
3. Avinash Kaushik., “Web Analytics 2.0, The Art of Online Accountability and Science of Customer” , Wiley

4. Michael Beasley, “Practical Web Analytics for User Experience”, Morgan Kaufmann
5. Stephan A. Miller, Piwik “Web Analytics Essentials”, Packt
6. Jim Sterne , “Web Metrics: Proven Methods for Measuring Web Site Success”, Wiley

Accomplishments of the student after completing the Course:

- Ability to decide and implement appropriate data model for WWW
- Ability to design customized search engines
- Ability to perform Web Analytics

Course Name: Machine Learning - Practical

Course Code: MCSDE423(2)

Objectives:

The main objective of this course is to enable students to

- Implement the theoretical knowledge of Machine Learning through programming
- Identify, formulate and solve machine learning problems that arise in practical applications
- Explore various tools and technologies that enable Machine Learning

Prerequisites:

- Mathematical Foundations, Programming in C++/Python

Contents:

1. Regression & K-Nearest Neighbor

Implementation of following concepts:

Linear Regression: Simple linear regression, Multiple linear regression, Model evaluation (R-squared, mean squared error), Regularization (Lasso, Ridge)

Logistic Regression: Sigmoid function, Binary classification using logistic regression, Multiclass classification using logistic regression, Model evaluation (confusion matrix, accuracy, precision, recall, F1score), Regularization (Lasso, Ridge)

K-Nearest Neighbors: Distance metrics (Euclidean, Manhattan, Cosine), KNN algorithm for classification and regression, Model evaluation (cross-validation)

2. Linear Regression & Logistic Regression

Implementation of following concepts:

Decision Trees: Information gain and entropy, ID3 and C4.5 algorithms for decision tree construction, Pruning decision trees, Model evaluation (confusion matrix, accuracy, precision, recall, F1-score)

Random Forests: Bagging, Construction of random forests for classification and regression, Model evaluation (out-of-bag error, feature importance)

Support Vector Machines: Maximum margin classifier, soft margin classifier, Non-linear SVM using kernel methods, Model evaluation (confusion matrix, accuracy, precision, recall, F1-score)

Unsupervised Learning: K-means clustering, Hierarchical clustering, Principal component analysis, Anomaly detection

Ensemble Methods: Boosting (AdaBoost), Bagging (Random Forest), Stacking

References:

1. Mitchell T., “Machine Learning”, McGraw Hill Publications
2. Grus J,. “Data Science from Scratch”, O’Reilly Publications
3. Alpaydin E., “Introduction to Machine Learning”, MIT Press
4. Coelho R., “Building Machine Learning Systems with Python”, Packt Publishing Ltd.
5. Marsland S., “MACHINE LEARNING: An Algorithmic Perspective, CRC Press
6. Wunsch D,. Xu R,. “Clustering”, IEEE Press
7. “Introduction to Machine Learning”; Ethem Alpaydın; The MIT Press

Accomplishments of the student after completing the course:

After completion of the course, students should be able to:

- Implement various machine learning algorithms
- Understand how to evaluate models generated from data
- Develop application involving computer vision and Natural Language Processing

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Course Name: Deep Learning - Practical

Course Code: MCSDE424(2)

Objectives:
The aim of this course is to enable students to
- Introduce with the latest algorithms and architectures of deep learning with practical viewpoint
- The necessary background to understand the ongoing research and gain required implementation knowledge

Prerequisites:
Basics of Linear Algebra, Calculus, Probability and Machine Learning Fundamentals

Contents:

1. **Deep Learning Frameworks**
   Overview & Installation of deep learning frameworks like TensorFlow, Keras, PyTorch
   Implementing Artificial neural networks (ANN), Activation functions (sigmoid, ReLU, tanh), Gradient descent optimization, Backpropagation algorithm

2. **Convolutional Neural Networks, Recurrent and Recursive Nets**
   Convolutional Neural Networks (CNN)
   Convolution operation, Pooling layer, Image classification using CNN
   Recurrent Neural Networks (RNN)
   Sequence data and LSTM networks, Text classification using RNN, Language generation using RNN

   Using pre-trained models for new tasks, Fine-tuning pre-trained models, Transfer learning for computer vision tasks

References:

1. Ian Goodfellow, Yoshua Bengio and Aaron Courville, "*Deep Learning*", MIT Press
2. Jeff Heaton, "*Artificial Intelligence for Humans: Deep Learning and Neural Networks Book – 3*", Heaton Research Inc.
3. Giancarlo Zaccone, "*Deep Learning with Tensor Flow*", Packt Publishing Ltd.

Accomplishments of the student after completing the Course:
At the end of the work students will be able to

- Understand various algorithms for deep learning
- Understand the concepts and architecture related to deep learning
- Design models to solve real life challenges using deep learning

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Course Name: Blockchain Technology - Practical

Course Code: MCSDE423(3)

Objectives:

The aim of this course is

- To implement blockchain technology and Cryptocurrency using programming
- Students can integrate ideas from blockchain technology into their own projects.

Prerequisites: Database Management System

Contents:

1. **Working with Ethereum**
   Ethereum's Four Stages of Development, Ethereum: A General-Purpose Blockchain Ethereum's Components, From General-Purpose Blockchains to Decentralized Applications, DApps)

   Ethereum Basics
   Ether Currency Units, Choosing an Ethereum Wallet, Control and Responsibility, Getting Started with MetaMask, Creating a Wallet, Switching Networks, Getting Some Test Ether, Sending Ether from MetaMask, Exploring the Transaction History of an Address, Introducing the World Computer, Externally Owned Accounts (EOAs) and Contracts
   A Simple Contract: A Test Ether Faucet, Compiling the Faucet Contract, Creating the Contract on the Blockchain, Interacting with the Contract, Viewing the Contract Address in a Block Explorer, Funding the Contract, Withdrawing from Our Contract

   Ethereum Clients
   Ethereum Networks, Running a full Node, Full Node Advantages and Disadvantages, Public Testnet Advantages and Disadvantages, Local Blockchain Simulation Advantages and Disadvantages, Running an Ethereum Client, Hardware Requirements for a Full Node, Software Requirements for Building and Running a Client (Node), Parity, Go-Ethereum (Geth), The First Synchronization of Ethereum-Based Blockchains, Running Geth or Parity, The JSON-RPC Interface, Remote Ethereum Clients, Mobile (Smartphone) Wallets, Browser Wallets, Ethereum's Cryptographic Hash Function, Ethereum Addresses, Ethereum Address Formats, Inter Exchange Client Address Protocol, Hex Encoding with Checksum in Capitalization (EIP-55)

2. **Wallets, Transactions & Smart Contracts:**
   Wallet Technology Overview, Nondeterministic (Random) Wallets, Deterministic (Seeded) Wallets, Hierarchical Deterministic Wallets (BIP-32/BIP-44), Seeds and Mnemonic Codes (BIP-39), Wallet Best Practices, Mnemonic Code Words (BIP-39), Creating an HD Wallet from the Seed, HD Wallets (BIP-32) and Paths (BIP-43/44)

The Structure of a Transaction, The Transaction Nonce, Keeping Track of Nonces, Gaps in Nonces, Duplicate Nonces, and Confirmation, Concurrency, Transaction Origination, and Nonces, Transaction Gas, Transaction Recipient, Transaction Value and Data, Transmitting Value to EOAs and Contracts, Transmitting a Data Payload to an EOA or Contract, Special Transaction: Contract Creation, Digital Signatures, The Elliptic Curve Digital Signature Algorithm, How Digital Signatures Work, Verifying the Signature ECDSA Math, Transaction Signing in Practice, Raw Transaction Creation and Signing Raw Transaction Creation with EIP-155, The Signature Prefix Value (v) and Public Key Recovery, Separating Signing and Transmission (Offline Signing), Transaction Propagation, Recording on the Blockchain, Multiple-Signature (Multisig) Transactions

Overview of Smart Contracts, Solidity, Vyper, Tokens, Oracles, Building Decentralized Apps with Ethereum

References:

1. Antonopoulos A., Wood G, "Mastering Ethereum", O'Reilly Media
   Bauer Davi, "Getting Started with Ethereum", Apress
   "Blockchain Technology", by Chandramouli Subramanian, Asha A George, Abhilash K A and Meena Karthikeyan, Universities Press
2. "Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction", by Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller and Steven Goldfeder, Princeton University Press
3. "Blockchain Blue print for Economy" by Melanie Swan
4. "Blockchain Basics: A Non-Technical Introduction in 25 Steps", by Daniel Drescher
5. " Mastering Blockchain", by Imran Bashir, 2/E, Packt publishing, Mumbai.

Accomplishments of the student after completing the Course:

- Students should be able to understand Blockchain Mechanism.
- Students should be able to understand Various Applications of Blockchain.

Course Name: Internet of Things - Practical

Course Code: MCSDE424(3)

Objectives:
The aim of this course is to enable students to

- Understand general concepts of Internet of Things (IoT)
- Recognize various devices, sensors and applications
- Apply design concept to IoT solutions
- Analyze various M2M and IoT architectures
- Evaluate design issues in IoT applications
- Create IoT solutions using sensors, actuators and Devices

Prerequisites:
Basics of Wireless networking, Network Security

Contents:

1. **Programming the IoT**

   Setting Up Your Development and Test Environment, Testing the Constrained Device App, Testing the Gateway Device App, Design Concepts, Add System Performance Tasks to the Constrained Device Application, Add System Performance Tasks to the Gateway Device Application

   Simulating Sensors and Actuators, Generating Simulated Data Using a Sensor Data Generator Class, Integrating Sensing and Actuation Simulation Within Your Application Design, Representing Sensor and Actuator Data Within the Application, Create Data Containers to Support Data Collection and Actuation, Simulating Sensors
   Simulating Actuators, Connecting Simulated Sensors with the Sensor Adapter Manager
   Connecting Simulated Actuators with the Actuator Adapter Manager, Create and Integrate the Device Data Manager, Hysteresis Management

   Emulating Sensors and Actuators, Setting Up and Configuring an Emulator, The Sense-Emu Sense HAT Emulator, Integrating Sensing and Actuation Emulation Within Your Application Design, Emulating Sensors, Emulating Actuators, Connecting Emulated Sensors with the Sensor Adapter Manager, Connecting Emulated Actuators with the Actuator Adapter Manager, Threshold Management

   Data Translation and Management Concepts, Data Translation in the Constrained Device App, Data Translation in the Gateway Device App, Proactive Disk Utilization Management

2. **IoT Integration to Objects**

   MQTT Integration, About MQTT, Connecting to a Broker, Message Passing, Control Packets and the Structure of an MQTT Message, Adding MQTT to Your Applications

Installing and Configuring an MQTT Broker, Create the MQTT Connector Abstraction Module, Add Callbacks to Support MQTT Events, Add Publish, Subscribe, and Unsubscribe Functionality, Integrate the MQTT Connector into Your CDA, Create the MQTT Connector Abstraction Module, Add Callbacks to Support MQTT Events, Add Publish, Subscribe, and Unsubscribe Functionality, Integrate the MQTT Connector into Your GDA, Security and Overall System Performance, Subscriber Callbacks, CDA to GDA Integration

CoAP Server Implementation, About CoAP, Client to Server Connections, Request Methods, Message Passing, Datagram Packets and the Structure of a CoAP Message, Add CoAP Server Functionality to the Gateway Device Application, Add CoAP Server Functionality to the Constrained Device Application, Add More Resource Handlers, Add a Custom Discovery Service, Add Dynamic Resource Creation

CoAP Client Integration, Add CoAP Client Functionality to the Constrained Device Application, Add CoAP Client Functionality to the Gateway Device App, Add a Robust OBSERVE Cancel Feature, Add Support for DELETE and POST

Edge Integration, Adding TLS Support to MQTT Broker, Add Security Features to Gateway Device App MQTT Client Connector, Add Security Features to Constrained Device App MQTT Client Connector, Adding Business Logic to the Gateway Device App, Adding Business Logic to the Constrained Device App, Add DTLS Support to Your CoAP Client and Server, Integrating IoT with Cloud Services

References:

1. Kind A., “Programming the Internet of Things”, O’Reilly Media
2. Vijay Madisetti and Arshdeep Bahga, “Internet of Things (A Hands-on-Approach)” ,1st edition, Orient Blackswan, Universities Press
3. Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting everything”, 1st edition, Apress Publications
4. Cuno Pfister, “Getting Started with the Internet of Things”, O’Reilly Media, 2011, ISBN: 978-1-4493-357-1

Accomplishments of the student after completing the Course:

At the end of the work students will be able to

- Apply IoT concepts and IoT Standards
- Understand Components and relevance of IoT System for the future
- Build IoT Applications
- Apply IoT in smart city environment in Indian scenario
- Analyze challenges in IoT implementation
